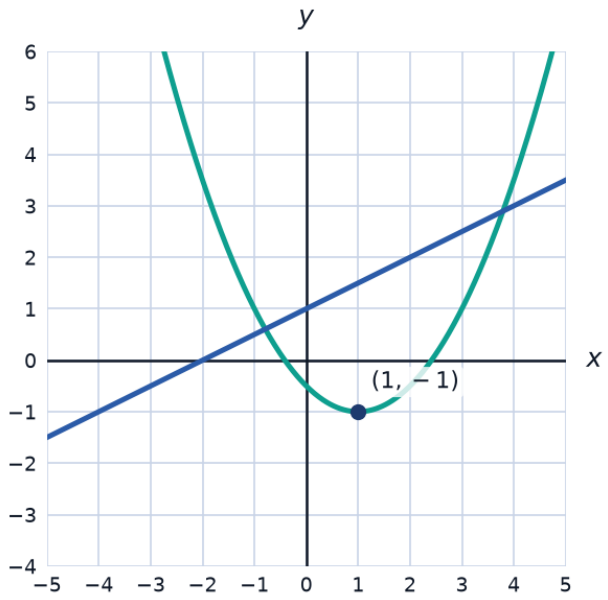
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

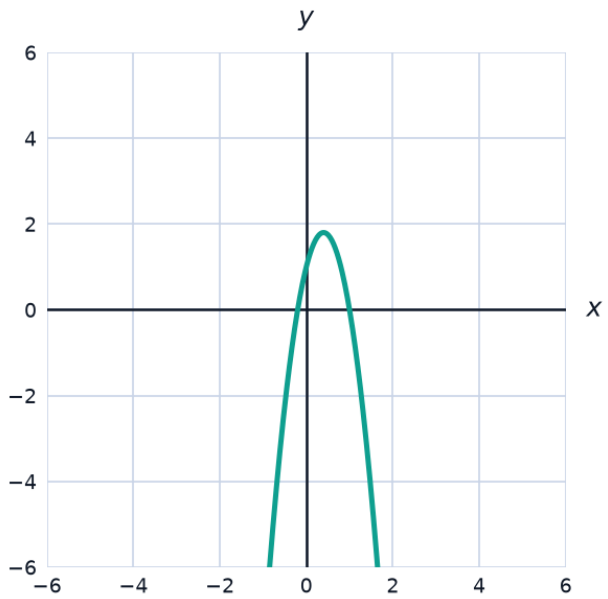
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve $\frac{3}{x-1} + 2 = 5$.



2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

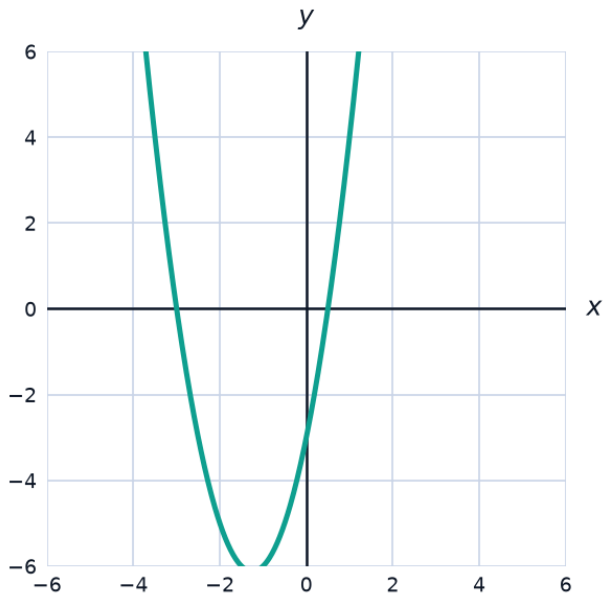
Find the slant (oblique) asymptote of the rational function: $g(x) = \frac{2x^3 - 5x^2 + 4x + 1}{x^2 - x + 1}$



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3 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

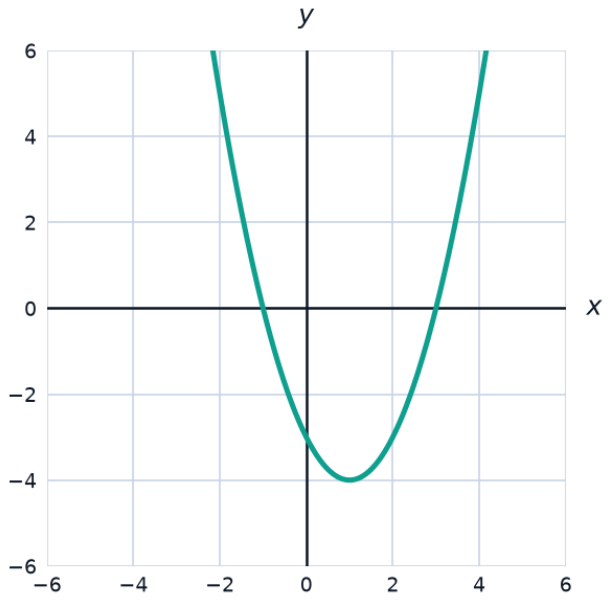
Find the vertical and horizontal asymptotes of the rational function: $f(x) = \frac{4x^2 - 1}{2x^2 + 5x - 3}$



4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the function $f(x) = (x - 1)^2 - 4$ is shown in the standard (x, y) coordinate plane below.

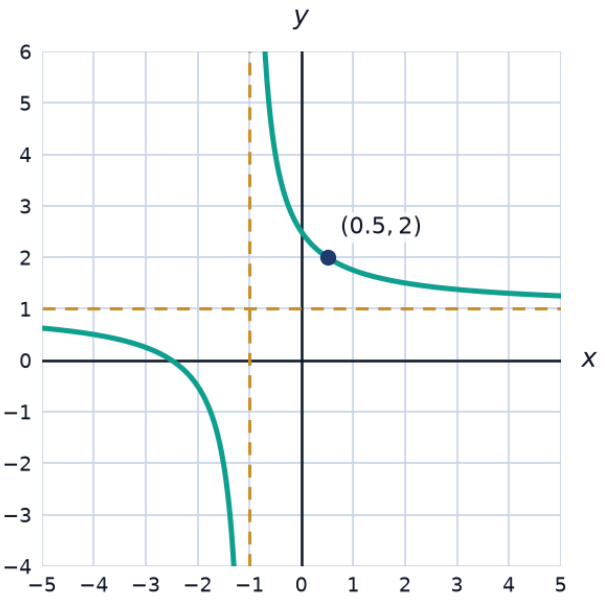
What is the range of f?



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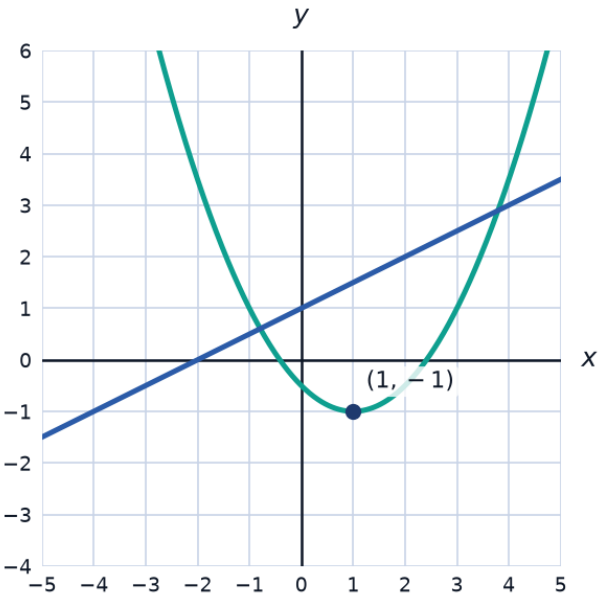
5 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Solve the rational equation: $\frac{2}{x-3} + \frac{1}{x} = \frac{x-1}{x-3}$



6 [Foundational] • Target: Function composition

Let $f(x) = 3x - 2$ and $g(x) = x^2 + 1$. Find $f(g(2))$ and $g(f(2))$.



7 [Foundational] • Target: Function composition

Using the same f and g, find the formula for $(f \circ g)(x)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Inverse functions

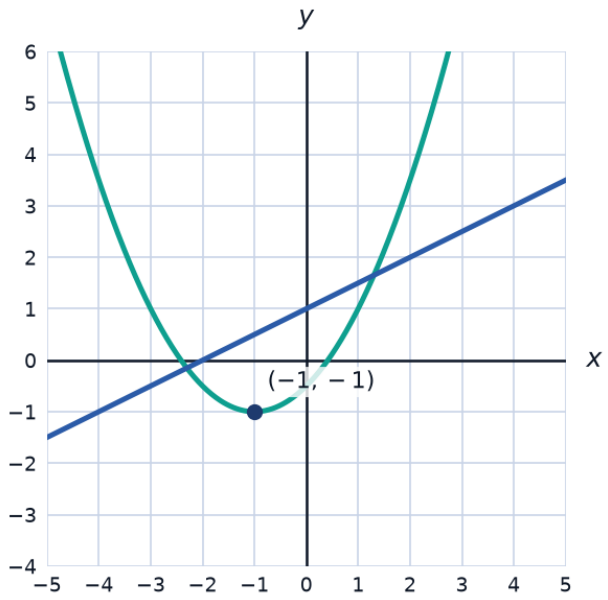
Find $f^{-1}(x)$ for $f(x) = 3x - 2$.

9 [Intermediate] • Target: Inverse functions

Verify your answer above by computing $f(f^{-1}(5))$.

10 [Intermediate] • Target: Rational equations

Solve $\frac{2x-1}{x+3} = \frac{1}{2}$.



11 [Intermediate] • Target: Rational equations

Solve $\frac{3}{x-1} + 2 = 5$.

12 [Intermediate] • Target: Factoring higher-degree polynomials

Given that $x = 1$ is a root of $x^3 - 2x^2 - 5x + 6 = 0$, use the Factor Theorem to factor completely and find all roots.

13 [Intermediate] • Target: Factoring higher-degree polynomials

Verify that $x = -2$ is a root of $x^3 - 2x^2 - 5x + 6 = 0$ by direct substitution.

14 [Intermediate] • Target: Exponential and logarithmic equations

Solve $3^{x-1} = 81$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

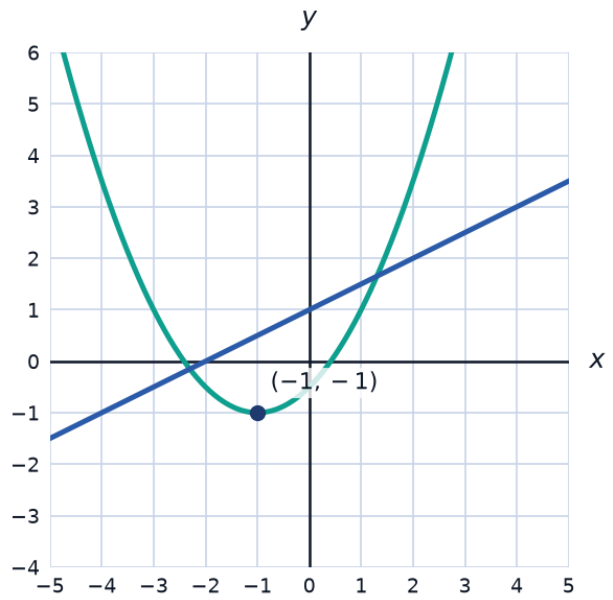
15 [Challenging] • Target: Exponential and logarithmic equations • Cross-Lesson: 3.01 Sequences & Series

Solve $\log_3(x+2) + \log_3(x-2) = 2$, checking for extraneous roots.

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16 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

Solve $\frac{2x-1}{x+3} = \frac{1}{2}$.



17 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Limits

Find the inverse function $f^{-1}(x)$ for $f(x) = \frac{2x+3}{x-5}$ where $x \neq 5$.

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Complex Numbers

Find all real solutions to the radical equation: $\sqrt{2x+15} - x = 0$

19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Matrices & Vectors

If r_1 and r_2 are the roots of $3x^2 - 8x + 5 = 0$, find the exact value of $\frac{1}{r_1} + \frac{1}{r_2}$ without solving for the roots individually.

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Sequences & Series

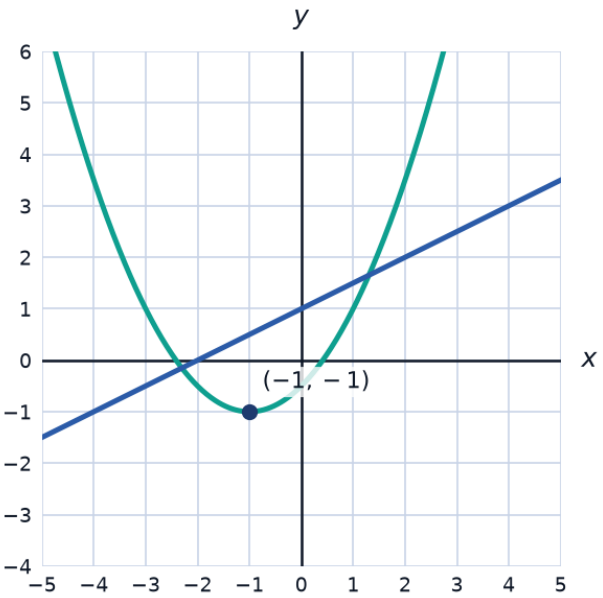
Convert the repeating decimal $0.\overline{45}$ to an irreducible common fraction.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve $\frac{4x-3}{x-8}=5$.



a. 38

b. 36

c. 37

d. −37

22 [Foundational] • Target: Inverse functions

Let $f(x)=x-9$ and $g(x)=-3x^2+7$. Find $f(g(0))$.

a. 0

b. 7

c. −4

d. −2

23 [Foundational] • Target: Rational equations

Let $f(x)=-2x-6$ and $g(x)=-x^2-1$. Find $f(g(-4))$.

a. 28

b. −17

c. 26

d. 30

24 [Foundational] • Target: Factoring higher-degree polynomials

Let $f(x)=x+7$ and $g(x)=x^2-8$. Find $f(g(-5))$.

a. 26

b. 24

c. 17

d. 22

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25 [Foundational] • Target: Exponential and logarithmic equations

Let $f(x) = 4x + 4$ and $g(x) = -3x^2 - 7$. Find $f(g(-4))$.

a. -218

b. -55

c. -214

d. -216

26 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Let $f(x) = -3x + 3$ and $g(x) = -3x^2 + 3$. Find $f(g(1))$.

a. 1

b. 0

c. 3

d. 5

27 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Let $f(x) = -4x - 9$ and $g(x) = -4x^2 - 3$. Find $f(g(-3))$.

a. -39

b. 145

c. 147

d. 149

28 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Let $f(x) = 2x + 3$ and $g(x) = 4x^2 - 8$. Find $f(g(5))$.

a. 187

b. 185

c. 189

d. 92

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve $\frac{2x-4}{x+4} = -2$.

a. -1

b. 0

c. 1

d. -2

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Solve $\frac{3x+4}{x-9} = -2$.

a. $-\frac{14}{5}$

b. $\frac{14}{5}$

c. $\frac{9}{5}$

d. $\frac{19}{5}$

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Solve $\frac{x+9}{x-2} = -4$.

a. $-\frac{1}{5}$

b. $\frac{4}{5}$

c. $\frac{1}{5}$

d. $-\frac{6}{5}$

32 [Intermediate] • Target: Function composition

Solve $\frac{-4x+5}{x-2} = -3$.

a. 0

b. 1

c. -2

d. -1

33 [Intermediate] • Target: Inverse functions

Solve $\frac{-x+2}{x-3} = 3$.

a. $\frac{7}{4}$

b. $\frac{11}{4}$

c. $-\frac{11}{4}$

d. $\frac{15}{4}$

34 [Intermediate] • Target: Rational equations

Solve $\frac{-x+9}{x-1} = 2$.

a. $\frac{11}{3}$

b. $\frac{14}{3}$

c. $\frac{8}{3}$

d. $-\frac{11}{3}$

35 [Intermediate] • Target: Factoring higher-degree polynomials

Find $f^{-1}(x)$ for $f(x) = 2x - 5$.

a. $\frac{x+5}{2}$

b. $\frac{x+5}{-2}$

c. $2x - 5$

d. $\frac{x-5}{2}$

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36 [Intermediate] • Target: Exponential and logarithmic equations

Find $f^{-1}(x)$ for $f(x) = 4x + 7$.

a. $\frac{x-7}{-4}$

b. $4x + 7$

c. $\frac{x-7}{4}$

d. $\frac{x+7}{4}$

37 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Given that $x = 4$ is a root of $x^3 - 13x - 12 = 0$, use the Factor Theorem to find the other two roots.

a. $(-3, -1)$

b. $(3, 1)$

c. $(-2, -1)$

d. $(-3, -2)$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Complex Numbers

Given that $x = 3$ is a root of $x^3 - 10x^2 + 27x - 18 = 0$, use the Factor Theorem to find the other two roots.

a. $(2, 6)$

b. $(1, 5)$

c. $(-1, -6)$

d. $(1, 6)$

39 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Matrices & Vectors

Given that $x = 2$ is a root of $x^3 + 6x^2 - 9x - 14 = 0$, use the Factor Theorem to find the other two roots.

a. $(-6, -1)$

b. $(-7, -2)$

c. $(7, 1)$

d. $(-7, -1)$

40 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Sequences & Series

Solve $5^{2x+7} = 5$ for x.

a. -4

b. -2

c. -1

d. -3

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41 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

Solve $2^x = 8$ for x.

a. 5

b. 2

c. 3

d. 4

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Limits

Solve $\frac{3x-6}{x+1} = -4$.

a. $\frac{9}{7}$

b. $\frac{2}{7}$

c. $-\frac{5}{7}$

d. $-\frac{2}{7}$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Complex Numbers

Solve $\frac{4x+6}{x-7} = -4$.

a. $\frac{11}{4}$

b. $-\frac{11}{4}$

c. $\frac{15}{4}$

d. $\frac{7}{4}$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Matrices & Vectors

Solve the rational inequality: $\frac{x-3}{x+2} \leq 0$.

a. $(-2, 3]$

b. $[-2, 3]$

c. $(-\infty, -2) \cup [3, \infty)$

d. $x \leq 3$

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45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Sequences & Series

What is the form of the partial fraction decomposition for $\frac{3x+1}{(x-1)(x^2+4)}$?

a. $\frac{A}{x-1}+\frac{Bx+C}{x^2+4}$

b. $\frac{A}{x-1}+\frac{B}{x^2+4}$

c. $\frac{A}{x-1}+\frac{B}{x+2}+\frac{C}{x-2}$

d. $\frac{Ax+B}{x-1}+\frac{C}{x^2+4}$

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